In Memory of William D. Loughlin

Robert W. Loughlin had long sought a fitting way to honor the memory of his father, William D. Loughlin, a prominent radio engineer and founder of Boonton Radio Corporation. A pioneer in the development of radio, William Loughlin pursued his love of the new medium and rode the wave of research and development in that emerging industry through the first half of the 20th century. In establishing a charitable remainder unitrust to benefit the Schools of Natural Sciences and Historical Studies at the Institute for Advanced Study, Bob and Ginny Loughlin hope to encourage William Loughlin's spirit of devotion to learning and experimentation at the edges of knowledge in future generations of scientists and scholars.

Born in Philadelphia in 1893, William Loughlin was one of the early experimenters in radio. He built his first set at the age of fifteen, and continued his hobby while a student at St. Joseph's College, where he designed, built and installed a 5 kilowatt wireless station in 1911. Because of his work, the college was granted one of the first experimental licenses by the U.S. Government, authorizing transmission of wireless telegraphy.

The outbreak of World War I galvanized the government's interest in the new concept of using radio for communication and navigation. The sinking of the Titanic was the first major event to demonstrate the impact of radio, which made possible the summoning of the White Star ship, Carpathia, 58 miles away but also equipped with a Marconi radio. Within three hours of the sinking, the Carpathia arrived to rescue those survivors in lifeboats. Nearer ships, without radios and thus ignorant of the disaster, did nothing to help.

The U.S. Navy was at the forefront of the development of radio, and William Loughlin was tapped by his friend, Stuart Ballantine, at the Philadelphia Navy Yard's Laboratory, to join sensitive work in progress on a critical navigational device, the radio compass. In addition to its navigational role, this device proved crucial in intercepting radio signals between German ships and establishing their location. "This work," noted Bob Loughlin, "was the beginning of what is today known as 'elit' – electronic intelligence gathering."

With the departure of Stuart Ballantine after the war, William Loughlin stepped up to lead the Lab's work, and oversaw the continuing development of radio communication and navigation equipment for ships, aircraft and dirigibles, as well as special tasks such as the Navy's NC series of seaplanes that attempted the first trans-Atlantic crossing in 1919. The NC-4 was the sole aircraft to succeed.

Meanwhile, a series of inventions and developments in the area of plastics in the early 1900s combined to provide the materials needed for the expanding electrical and radio industries. Several key figures joined together to form the Radio Frequency Laboratories (RFL) to focus on using these new materials in solving the problems associated with the development of practical radio receivers. RFL's inventions in the field of broadcast receivers for the consumer market included single knob tuning, automatic volume control, and the ability of radios to utilize AC power rather than batteries. The company licensed its many patents to the various manufacturers of radios for the home.

William Loughlin, who joined RFL as a radio engineer in 1923, rose to become president in 1931. The company was dissolved in 1934 when the patents were sold to RCA, and the key personnel dispersed to follow their various interests, providing William the opportunity to turn his attention to the industry's pressing need for precision measuring and test instruments.

Up until this time, circuit development and testing had required radio engineers to employ many individual meters and involved complex calculations. In 1934, William Loughlin presented his newly-formed Boonton Radio Corporation's "Q-Meter" to the meeting of the Institute of Radio Engineers. The Q-Meter, which consolidated and simplified the testing process, was immediately adopted as an industry standard. With the advent of FM (frequency modulation) the Boonton Radio Corporation, in close cooperation with Major Edwin H. Armstrong, the inventor of FM, designed and produced the basic instruments needed by the radio and television industries.

The Second World War called for the development of many new electronic instruments. Boonton Radio worked closely with Bell Telephone Laboratories and the MIT Radiation Laboratories in developing the micro-wave signal generators needed for the development and maintenance of various kinds of radar equipment. For these and other outstanding contributions to the war effort, the Boonton Radio Corporation was awarded an Army-Navy "E" (for excellence) with four stars.

William Loughlin died in 1950, and the Boonton Radio Corporation was subsequently sold to the Hewlett-Packard Corporation, a young, upstart company heralding the "new new thing" on the electronics horizon. Today, radio has been eclipsed as a medium, but Bob Loughlin sees a fascinating parallel between the early, heady days of radio and today's ferment of creative endeavor surrounding the internet. "The analogy is incredibly strong. It all happened in a fairly short time, and then became the great, international rage. Radio had a sweeping influence – economic, philosophic, social – very much like the internet. And it was driven by a few visionary personalities who really made it happen."

Friends of the Institute, Bob and Ginny Loughlin take a serious, hands-on approach to philanthropy. Acknowledging that it was largely his father's hard work and vision that generated the resources that were passed on to him, Bob commented, "It is somewhat humbling to have the resources that we have because of my father. We feel a duty to share these in ways of which he would be proud." They have also encouraged the involvement of their five sons in this effort by setting up a family foundation to support interests identified by their sons and their families.

While there is obviously a natural connection between William Loughlin's career and support for physics at the Institute, Bob explained that his father was also a scholar in Greek and Latin, with a deep, personal love of history. For this reason, he structured the gift to honor his father to benefit the School of Historical Studies as well as the School of Natural Sciences. This tribute to the life and work of William D. Loughlin will serve to further the kind of in-depth scholarship and cutting-edge research that was so meaningful in his own life.

"My father had been involved in a field of exciting growth at an exciting time," said Bob. "I have a great respect for his being able to make meaningful technical contributions in a field that he loved, and the vision and courage to succeed." ■